

# Chapter 16

## Electric Fields and Potentials

# ELECTRIC FORCES, FIELDS, AND POTENTIALS

ELECTRIC FORCES

## **We will Discuss:**

- Electric Force **F** between two point charges. (Coulomb Law)
- Electric Field **E** of a point charge.
- Superposition Principle.
- Electric potential  $U$  and potential difference  $V$  or  $\Delta V$ , and Equipotential Surfaces.
- Motion of Charged particle in an Electric Field.
- Electric Field and Electric Potential for Charge Distribution: Sheet, Double Plates (Capacitor), and Conducting Sphere.
- Uniform Electric Field
- Electric Dipole
- Effects of Dielectrics.
- Capacitance and Energy Stored in a Capacitor.

## Electric Force: (between two point Charges)

If a charge  $q$  is at a distance  $r$  from a second charge  $Q$ , then The electric force on  $q$  (See Figure) is given by :

$$\vec{F} = k \frac{qQ}{r^2} \hat{r}$$

This is Coulomb's Law

$\hat{r}$  is a unit vector from  $Q$  to  $q$



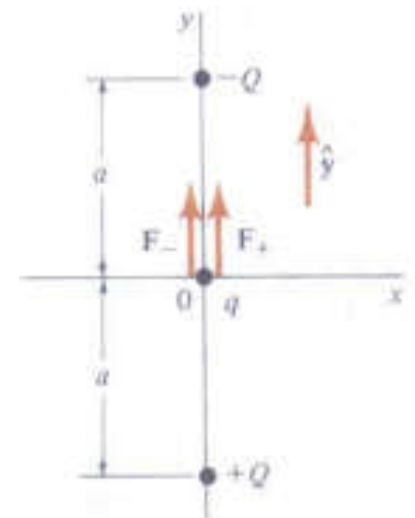
### Example 16.1

A positive charge  $q$  is near a positive charge  $Q$  and a negative charge  $-Q$  (Fig. 16.2). (a) Find the magnitude and direction of the force on  $q$ . (b) If  $q = 10^{-6}$  C,  $Q = 2 \times 10^{-6}$  C, and  $a = 1$  m, find the force on  $q$ .

$$\mathbf{F}_+ = \frac{kqQ}{a^2} \hat{\mathbf{y}}$$

$$\mathbf{F}_- = \frac{kq(-Q)}{a^2} (-\hat{\mathbf{y}}) = \frac{kqQ}{a^2} \hat{\mathbf{y}}$$

$$\mathbf{F} = \mathbf{F}_+ + \mathbf{F}_- = \frac{2kqQ}{a^2} \hat{\mathbf{y}}$$



$$\begin{aligned}
 \mathbf{F} &= \frac{2kqQ}{a^2} \hat{\mathbf{y}} \\
 &= \frac{2(9 \times 10^9 \text{ N m}^2 \text{ C}^{-2})(10^{-6} \text{ C})(2 \times 10^{-6} \text{ C})}{(1 \text{ m})^2} \hat{\mathbf{y}} \\
 &= 3.6 \times 10^{-2} \hat{\mathbf{y}} \text{ N}
 \end{aligned}$$

The force is directed upward.

# THE ELECTRIC FIELD

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Due to a point charge

$$\mathbf{F} = q_o \mathbf{E} \quad \longrightarrow \quad \mathbf{E} = \mathbf{F}/q_o$$

$$\vec{\mathbf{E}} = \frac{1}{q_o} \left[ k \frac{Q q_o}{r^2} \hat{\mathbf{r}} \right] = k \frac{Q}{r^2} \hat{\mathbf{r}}$$

For a point charge

$\mathbf{E}$  is a vector quantity  
With units N/C

For more than one point charge

$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots$  Vector Sum (**superposition Principle**)

### Example 16.2

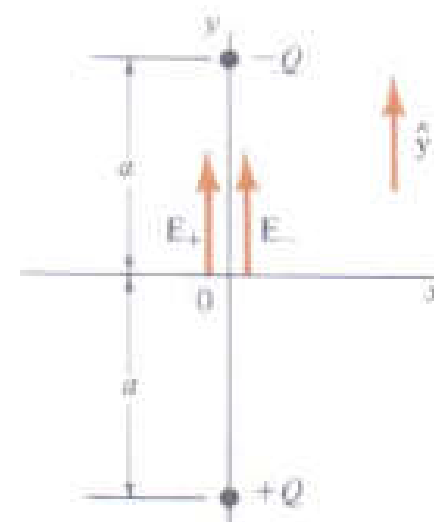
The charges  $Q$  and  $-Q$  of the preceding example are shown again in Fig. 16.6. (a) If  $Q = 2 \times 10^{-6} \text{ C}$  and  $a = 1 \text{ m}$ , find the electric field at the origin. (b) Find the force on the charge  $q = 10^{-6} \text{ C}$  placed at the origin.

(a) As before, the distance from the charge  $Q$  to the origin is  $a$ , and  $\hat{\mathbf{r}} = \hat{\mathbf{y}}$ . Using Eq. 16.4, the field at the origin due to this charge is

$$\mathbf{E}_+ = \frac{kQ}{a^2} \hat{\mathbf{y}}$$

$$\mathbf{E}_- = \frac{k(-Q)}{a^2} (-\hat{\mathbf{y}}) = \frac{kQ}{a^2} \hat{\mathbf{y}}$$

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = \frac{2kQ}{a^2} \hat{\mathbf{y}} = \frac{2(9 \times 10^9 \text{ N m}^2 \text{ C}^{-2})(2 \times 10^{-6} \text{ C})}{(1 \text{ m})^2} \hat{\mathbf{y}} = 3.6 \times 10^4 \hat{\mathbf{y}} \text{ N C}^{-1}$$





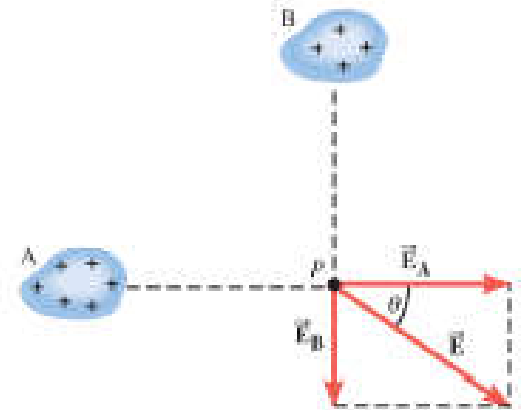
(b) The force on a charge  $q = 10^{-6} \text{ C}$  at the origin is

$$\begin{aligned}\mathbf{F} &= q\mathbf{E} = (10^{-6} \text{ C})(3.6 \times 10^4) \hat{\mathbf{y}} \text{ N C}^{-1} \\ &= 3.6 \times 10^{-2} \hat{\mathbf{y}} \text{ N}\end{aligned}$$

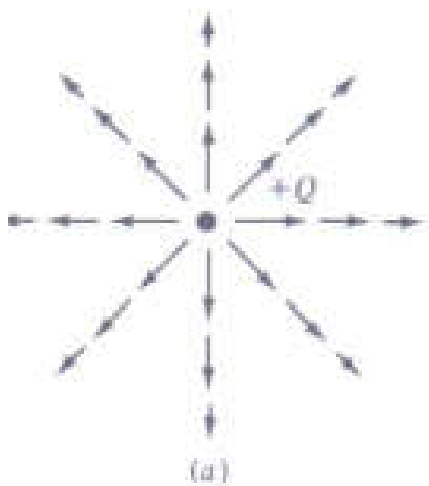
**Example:** The Figure shows two charged objects A and B. Each contributes electric field at point P:  $E_A = 3\text{ N/C}$  directed to the right, and  $E_B = 2\text{ N/C}$  directed downward. What is the net electric field at point P?

**Solution** The magnitude of the net electric field is

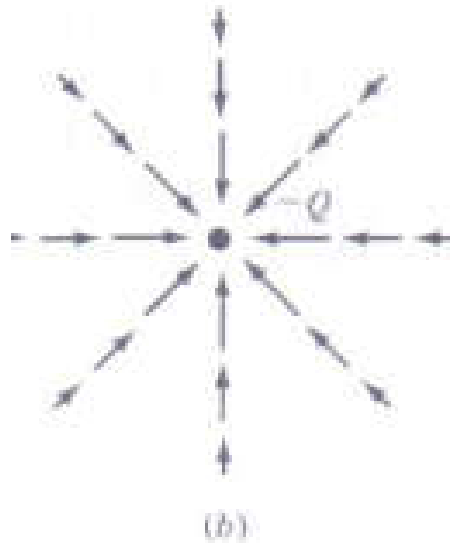
$$E = \sqrt{E_A^2 + E_B^2} = \sqrt{(3.00\text{ N/C})^2 + (2.00\text{ N/C})^2} = \boxed{3.61\text{ N/C}}$$



# Electric Field Diagrams

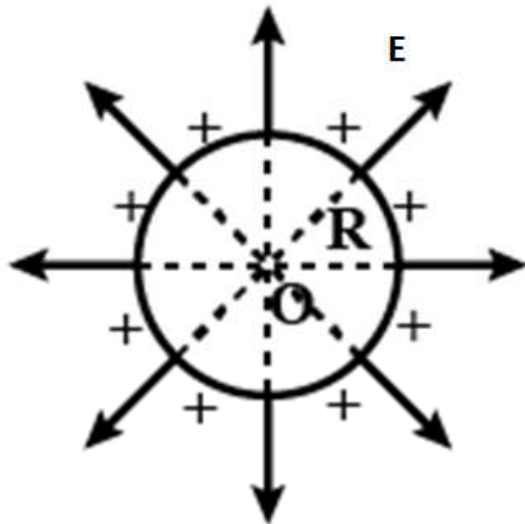


$\mathbf{E}$  for +ve Charge



$\mathbf{E}$  for -ve Charge

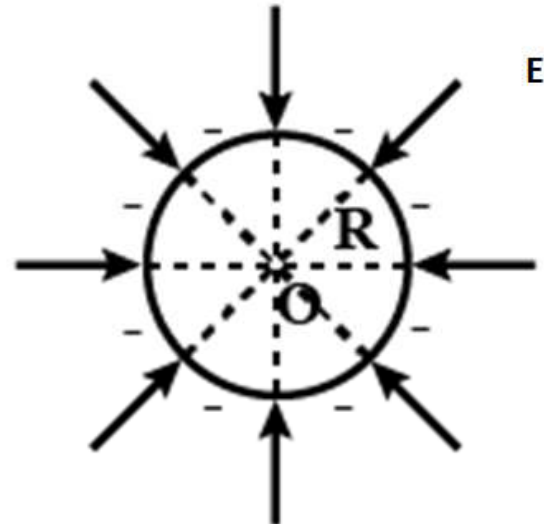
## Electric field lines for conducting sphere



**(a) Positively charged**

$$E(\text{inside}) = 0$$

$$E(\text{outside}) = kq/r^2$$



**(b) Negatively charged**

Surface charge density  $\sigma$

$$\sigma = Q/4\pi R^2$$

$R$  = radius of sphere

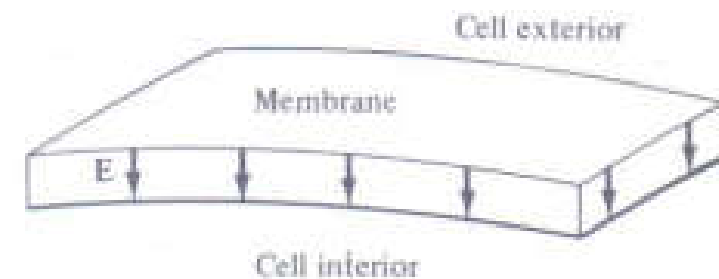
### Example 16.3

A thin, flat membrane separates a layer of positive ions outside a cell from a layer of negative ions inside (Fig. 16.17). If the electric field due to these charges is  $10^7 \text{ N C}^{-1}$ , find the charge per unit area  $Q/A$  in the layers on either side of the membrane.

Since the membrane is flat, the charges form uniformly charged planes on either side. Using  $E = 4\pi kQ/A$ , we have

$$\begin{aligned}\frac{Q}{A} &= \frac{E}{4\pi k} = \frac{10^7 \text{ N C}^{-1}}{4\pi(9 \times 10^9 \text{ N m}^2 \text{ C}^{-1})} \\ &= 88.4 \times 10^{-6} \text{ C m}^{-2} = 88.4 \mu\text{C m}^{-2}\end{aligned}$$

using  $1 \text{ microcoulomb} = 1 \mu\text{C} = 10^{-6} \text{ C}$ . Thus  $1 \text{ m}^2$  of membrane will have a net charge of  $+88.4 \mu\text{C}$  along the outside surface and a net charge of  $-88.4 \mu\text{C}$  on the inside.



## 16.4 | THE ELECTRICAL POTENTIAL

### and Motion of a charged Particle in an Electric Field

Suppose that at a certain position a charge  $q$  has an electrical potential energy  $\mathcal{U}$ . *Then the electric potential  $V$  at that position is defined to be the potential energy divided by the charge,*

$$V = \frac{\mathcal{U}}{q}$$

The unit of potential is the volt (V), where from this definition 1 volt = 1 joule per coulomb.

$$\begin{aligned} 1 \text{ eV} &= (1.60 \times 10^{-19} \text{ C})(1 \text{ V}) \\ &= 1.60 \times 10^{-19} \text{ J} \end{aligned}$$

### Example 16.4

In the cathode-ray tube of an oscilloscope or a television picture tube, electrons are accelerated from rest through a potential difference of  $+20,000$  V. What is their velocity? (The electron mass is  $9.11 \times 10^{-31}$  kg, and the charge is  $-e = -1.6 \times 10^{-19}$  C.)

From the definition of the electric potential, the change in potential energy is  $\Delta q\mathcal{U} = q \Delta V = (-e) \Delta V$ , where  $\Delta V = 20,000$  V. Then, from energy conservation,

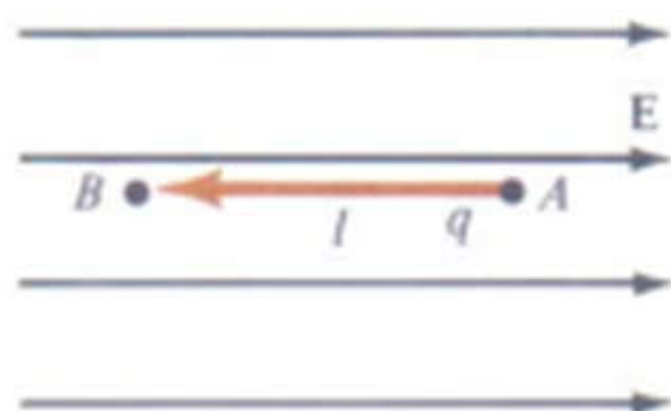
$$\frac{1}{2}mv^2 = e \Delta V$$

Hence

$$\begin{aligned}v &= \sqrt{\frac{2e \Delta V}{m}} \\&= \sqrt{\frac{2(1.60 \times 10^{-19} \text{ C})(20,000 \text{ V})}{9.11 \times 10^{-31} \text{ kg}}} \\&= 8.38 \times 10^7 \text{ m s}^{-1}\end{aligned}$$



We can find a relationship between the electric potential and field by considering a positive charge  $q$  in a constant electric field  $\mathbf{E}$



When a positive charge  $q$  is moved opposite to the field from  $A$  to  $B$ , its potential energy increases by  $\Delta \mathcal{U} = q \Delta V = qEl$ .

When the charge moves a distance  $l$  opposite to the field, the applied force does work  $Fl = qEl$ . Since the kinetic energy remains constant, this work must equal the change in the potential energy of the charge:

$$\Delta\mathcal{U} = qEl$$

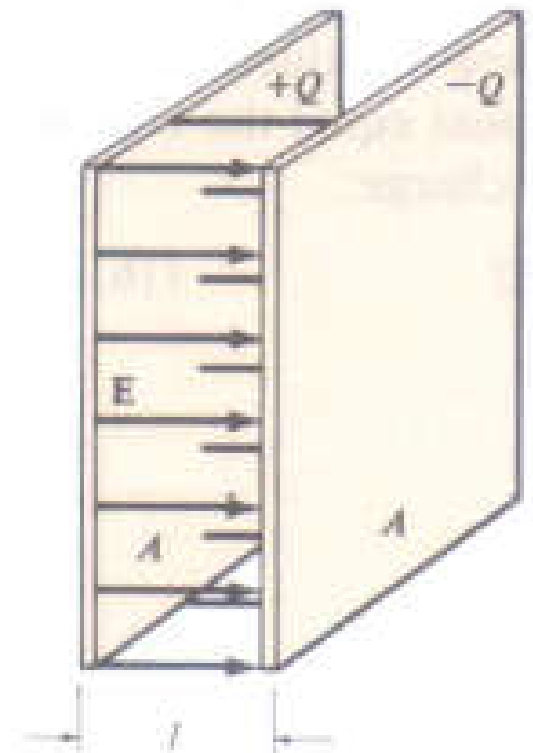
Dividing by  $q$ , we obtain the change in the electric potential,

$$\Delta V = El$$

the field

between two oppositely charged plates is uniform. If the plate area is  $A$  and the charges are  $+Q$  and  $-Q$ , the magnitude of the field is  $4\pi kQ/A$ . Thus when the plates are separated by a distance  $l$  (Fig. 16.19), the potential difference between them is  $\Delta V = El$ , or

$$\Delta V = \frac{4\pi kQ}{A} l \quad (\text{oppositely charged plates})$$



### Example 16.5

Two oppositely charged parallel plates have an area of  $1 \text{ m}^2$  and are separated by  $0.01 \text{ m}$ . The potential difference between the plates is  $100 \text{ V}$ . Find (a) the field between the plates; (b) the magnitude of the charge on a plate.

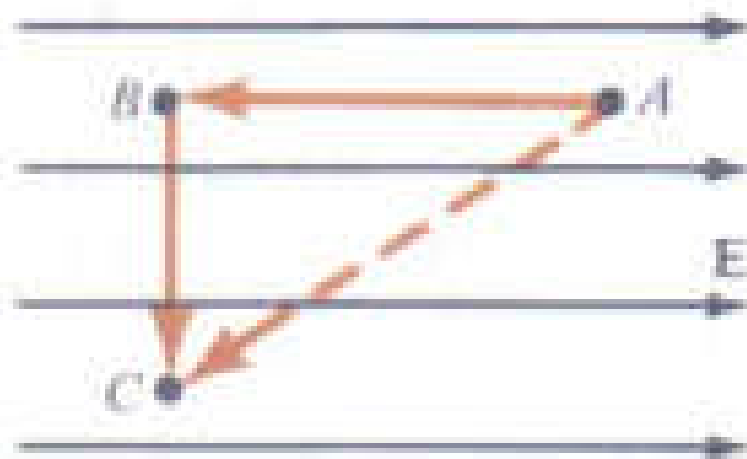
(a) Since the field is uniform,  $\Delta V = El$  can be used, and

$$E = \frac{\Delta V}{l} = \frac{100 \text{ V}}{0.01 \text{ m}} = 10^4 \text{ V m}^{-1}$$

(b) Using  $\Delta \mathcal{U} = qEl$  the charge on a plate has a magnitude

$$\begin{aligned} Q &= \frac{A \Delta V}{4\pi kl} = \frac{(1 \text{ m}^2)(100 \text{ V})}{4\pi(9 \times 10^9 \text{ N m}^2 \text{ C}^{-2})(10^{-2} \text{ m})} \\ &= 8.84 \times 10^{-8} \text{ C} \end{aligned}$$

## Note:



The work done by the conservative electric force is the same for the paths  $ABC$  and  $AC$ .

# The Potential Due to a Point Charge I

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The Potential energy for two point charges  $Q$  and  $q$  separated is

$$u = k \frac{Qq}{r}$$

Since  $V = U/q$  Then,

$$V = k \frac{Q}{r} \text{ (Point Charge)}$$

This expression implies that the electric potential  $V$  has been chosen to be zero at  $r = \infty$

### Example 16.6

A lead nucleus contains 82 protons in a spherically symmetric charge distribution with a radius of  $7 \times 10^{-15}$  m. A proton is fired from an accelerator directly at a lead nucleus. The initial velocity of the proton is  $10^7$  m s<sup>-1</sup>. How close will it come to the nucleus? (Assume that the nucleus is held stationary and that only electrical forces act between the proton and the nucleus.)



At a distance  $r$  from the center of the nucleus, since its charge is  $Q = +82e$ , its potential is

$$V = \frac{kQ}{r} = \frac{82ke}{r}$$

The potential energy of the proton is  $\mathcal{U} = qV = e(82ke)/r$ . At infinity, the potential energy is zero; at the point of closest approach, the kinetic energy is zero. Since the total energy is constant,

$$\frac{mv^2}{2} = \frac{82ke^2}{R}$$

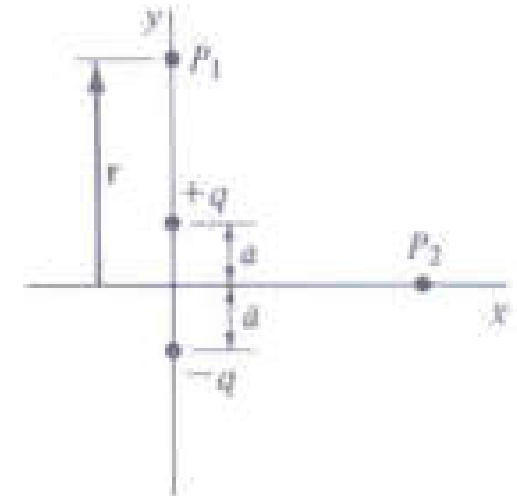
Solving for  $R$ , we have

$$\begin{aligned} R &= \frac{164ke^2}{mv^2} \\ &= \frac{164(9 \times 10^9 \text{ N m}^2 \text{ C}^{-2})(1.6 \times 10^{-19} \text{ C})^2}{(1.673 \times 10^{-27} \text{ kg})(10^7 \text{ m s}^{-1})^2} \\ &= 2.26 \times 10^{-13} \text{ m} \end{aligned}$$

This is about 30 times larger than the nuclear radius, which is  $7 \times 10^{-15} \text{ m}$ .

### Example 16.7

For the dipole in Fig. , find the electric potential at (a) point  $P_1$  on the  $y$  axis; (b) at point  $P_2$  on the  $x$  axis. (c) How much work is required to move a charge  $q$  from infinitely far away to point  $P_1$  if  $r = 3a$ ? (d) How much work is needed to move  $q$  from infinity to  $P_2$ ?



(a) At  $P_1$ , the potentials due to the two charges are

$$V_+ = \frac{kq}{r-a}, \quad V_- = \frac{-kq}{r+a}$$

The total electric potential is then

$$V = V_+ + V_- = kq \left[ \frac{1}{r-a} - \frac{1}{r+a} \right] = \frac{2kqa}{r^2 - a^2}$$

If  $r$  is large compared to  $a$ , then  $a$  can be neglected in the denominator, so

$$V = \frac{2kqa}{r^2}$$

(b) At point P2,  $V = 0$

(c) Using the result for  $V$  in part (a) with  $r = 3a$ , the potential at  $P_1$  is

$$V = \frac{2kqa}{(3a)^2 - a^2} = \frac{2kqa}{8a^2} = \frac{kq}{4a}$$

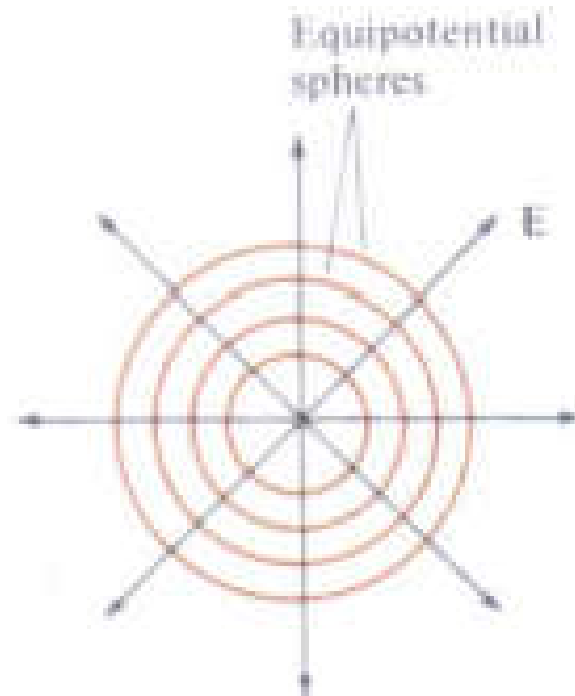
NOTE:

the potential energy of the system consisting of the dipole and the charge  $q$  is  $\mathcal{U} = qV = kq^2/4a$ . This is the amount of work needed to move  $q$  from infinity to  $P_1$ .

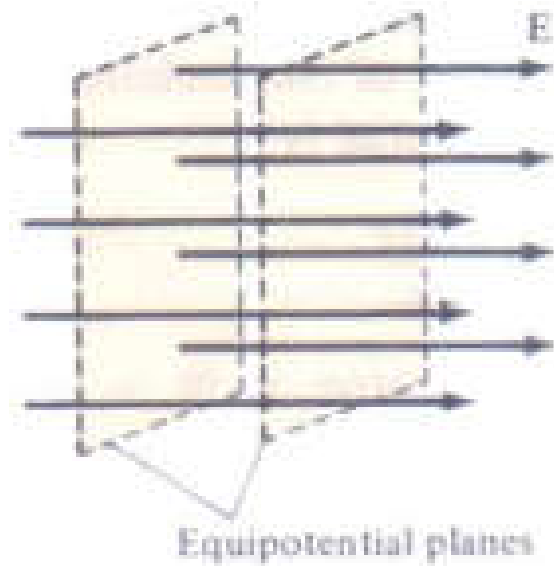
(d) Since  $V = 0$  at point  $P_2$ ,  $\mathcal{U} = qV = 0$ , and no work is needed to move the charge from infinity to  $P_2$ .

## 16.5 | EQUIPOTENTIAL SURFACES

(a) Point Charge: Equipotential surfaces for (a) a point charge; (b) a uniform electric field; (c) a dipole. Note that the equipotential surfaces and the electric field lines are always mutually perpendicular.



(b) Equipotential surfaces  
For uniform field  
(Capacitor)

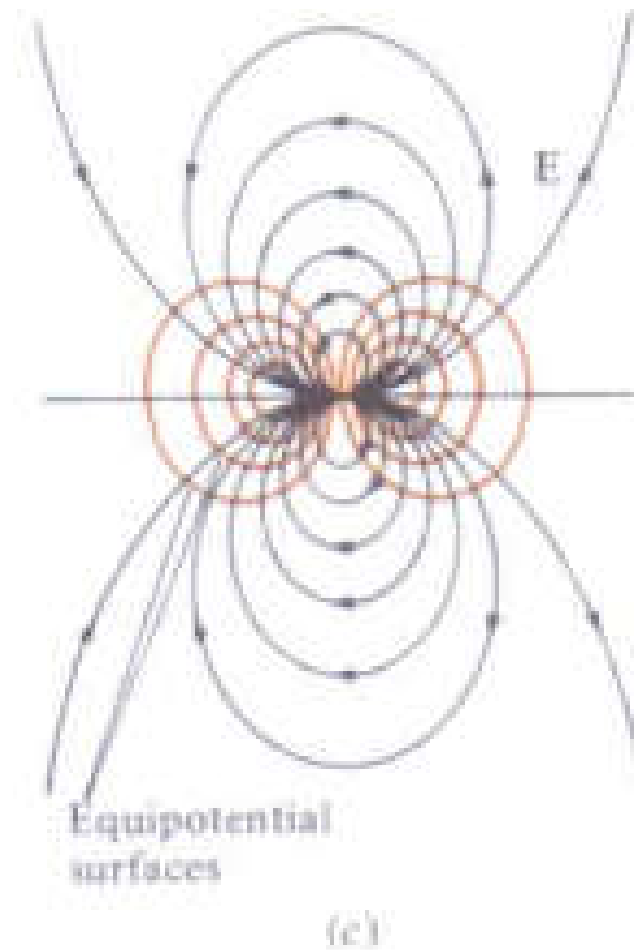


(b)



### (c) Equipotential Surfaces for Electric Dipole

What is the electric dipole?



# 16.6 Electric Dipole

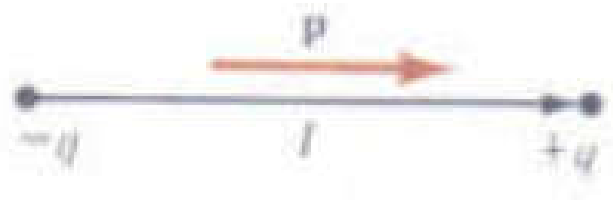
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An electric dipole consists of equal and opposite charges,  $+q$  and  $-q$ . The electric dipole moment is  $\mathbf{p} = ql$ .

$$\mathbf{p} = ql$$

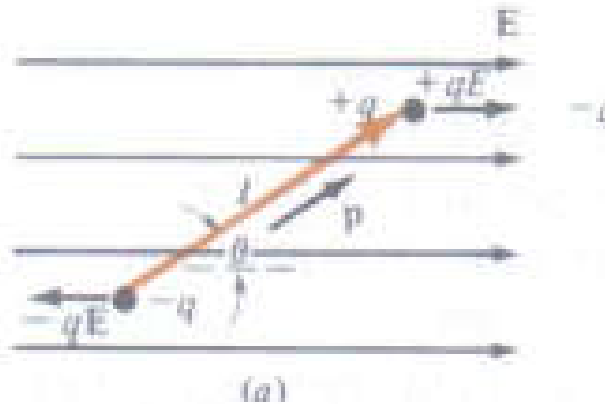
# 16.6 Electric Dipole



An electric dipole consists of equal and opposite charges  $+q$  and  $-q$ . The electric dipole moment is  $\vec{P} = q\vec{L}$

NOTE: In Example 16.7 the electric dipole is  $P = q(2a)$ .  
Check slide # 26

## Torque on an Electric Dipole placed in External Electric Field

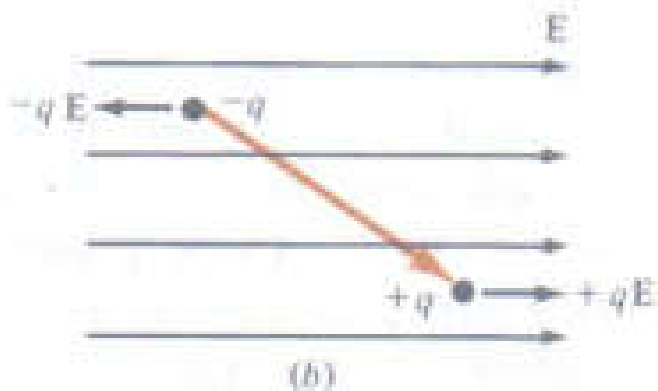


Torque on a dipole:

$$\boldsymbol{\tau} = \mathbf{l} \times (q\mathbf{E}) = \mathbf{p} \times \mathbf{E}$$

Potential Energy

$$U = -\mathbf{p} \cdot \mathbf{E}$$



## 16.8 | CAPACITANCE

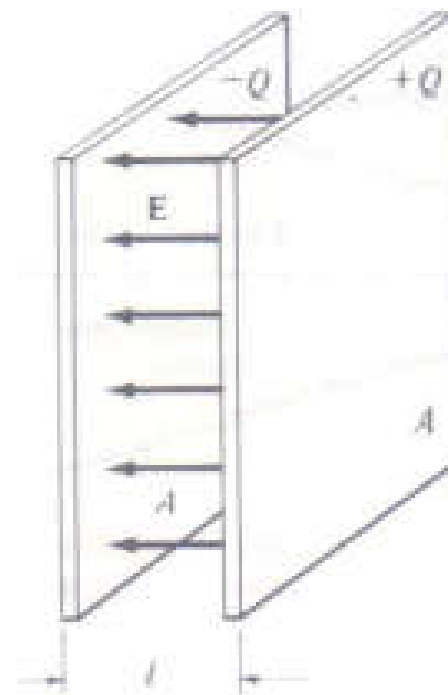
### The Parallel Plate Capacitor

$$C = \frac{Q}{V}$$

The unit of capacitance is the *farad* (F);

$$C = \frac{A}{4\pi kl}$$

$$C = \frac{\epsilon_0 A}{l}$$



A parallel plate capacitor.

### Example 16.10

A  $1\text{-}\mu\text{F}$  capacitor is connected to a  $12\text{-V}$  battery.  
What are the charges on its plates?

$$Q = CV = (10^{-6} \text{ F})(12 \text{ V}) = 1.2 \times 10^{-5} \text{ C}$$

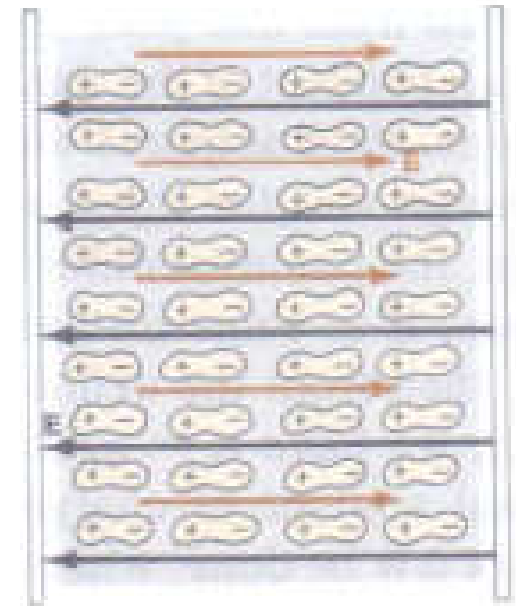
## 16.9 | EFFECTS OF DIELECTRICS

### Dielectric Materials

$$E_{\text{eff}} = E - E'$$

$$V = V_o/K, \quad K = \text{Dielectric constant}$$

$$C = \frac{K\epsilon_0 A}{l} \quad \text{Capacitance with Dielectric}$$



**TABLE 16.1**

Dielectric constants of several common insulators. Note that for air we can usually use  $K = 1$ , the value for a vacuum.

| Material                        | Temperature,<br>°C | Dielectric<br>Constant, $K$ |
|---------------------------------|--------------------|-----------------------------|
| Air (dry, at 1 atm)             | 20                 | 1.00059                     |
| Glass                           | 25                 | 5–10                        |
| Water                           | 25                 | 78                          |
|                                 | 80                 | 61                          |
| Plastics                        | 20                 | 3–20                        |
| Titanium dioxide                | 20                 | 100                         |
| Axon membrane<br>(unmyelinated) | 37                 | 8                           |
| Paper                           | 20                 | 3.5                         |



### Example 16.11

A capacitor is made of two foils, each of surface area  $1 \text{ m}^2$ , separated by paper  $0.05 \text{ mm} = 5 \times 10^{-5} \text{ m}$  thick. What is its capacitance?

According to Table 16.1,  $K = 3.5$  for paper. Thus the capacitance is

$$\begin{aligned} C &= \frac{K\epsilon_0 A}{l} \\ &= \frac{(3.5)(8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2})(1 \text{ m}^2)}{5 \times 10^{-5} \text{ m}} \\ &= 6.19 \times 10^{-7} \text{ F} = 0.619 \mu\text{F} \end{aligned}$$

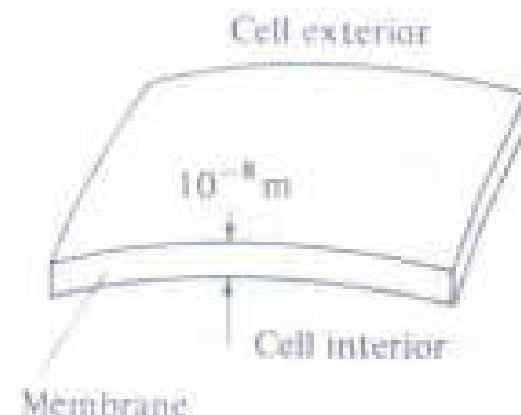
### Example 16.12

The ions inside and outside a cell are separated by a flat membrane  $10^{-8}$  m thick with a dielectric constant  $K = 8$ . Find the capacitance of  $1 \text{ cm}^2$  of membrane (Fig. 16.31).

For an area  $A = 1 \text{ cm}^2 = 10^{-4} \text{ m}^2$ , the capacitance is

$$C = \frac{K\epsilon_0 A}{l} = \frac{8(8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2})(10^{-4} \text{ m}^2)}{10^{-8} \text{ m}}$$

$$= 7.08 \times 10^{-7} \text{ F} = 0.708 \mu\text{F}$$



## 16.10 | ENERGY STORED IN A CAPACITOR

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### Parallel Plates Capacitor

$$U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$$

#### Example 16.13

One square centimetre of membrane has a capacitance of  $7.08 \times 10^{-7}$  F. If the potential difference across the membrane is 0.1 V, find the electrical energy stored in 1 cm<sup>2</sup> of membrane.

$$\begin{aligned} U &= \frac{1}{2} CV^2 = \frac{1}{2} (7.08 \times 10^{-7} \text{ F})(0.1 \text{ V})^2 \\ &= 3.54 \times 10^{-9} \text{ J} \end{aligned}$$